Rational Number Project

Level 2 / Lesson 23 / Overview

Students explore the meaning of multiplying two fractions less than one by concretely modeling story problems for fraction multiplication.

Materials

- Chips for students and teacher
- Student Page A

Teaching Actions

1. Present this story for students to model with chips.

   Twelve students are members of our after-school club. Two-thirds of the members are girls. One-fourth of the girls are also on the girls' track team. What fraction of the club is on the girls' track team?

2. Before solving, students should determine a reasonable range for the answer.

   a) Is the answer greater or less than 2/3 of the club?

   b) Is the answer greater or less than 1/2 of the club? Greater or less than 1/4 of the club?

3. Step back and let students work cooperatively on a solution.

Comments

1. Multiplication of fractions is a deceptively easy skill for students to learn. Answers to fraction multiplication can be obtained without considering the meaning of the quantities operated on or the meaning of the answer.

2. Fraction multiplication is introduced here in story contexts and students are asked to draw pictures or model with chips the actions in the stories.

3. You may want to break this lesson into two parts: story problems that use chips as a model and story problems that use pictures as a model.

Teaching Actions

4. Hints to use as needed:

   a) Use 12 chips to model 2/3.

   b) Perhaps organizing the chips as arrays will help.

   c) Are you considering 1/4 of the whole club or just part of the club?

   d) Is it possible for a certain number of chips to have more than one fraction name? Have you ever considered that 2 chips can have more than one fraction name?

5. Ask students to model their responses. Show how organizing the chips into arrays makes the problem easier to do. Compare answer to their initial estimates. Is the answer reasonable? Why would they expect the answer to be less than either fraction?

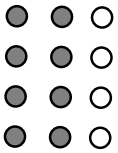
6. Record verbally the action in the story as:

   "One-fourth of 2-thirds is 2-twelfths or 1-sixth"

Comments

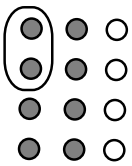
3. Note the concepts applied in the solution of the story problem:

   [Show 2/3 using 12 chips.]

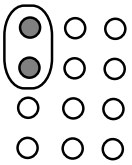
   2/3

   [Redefine the unit to be 8 chips and show 1/4 with 8 chips.]

   1/4 of 2/3

   [Re-evaluate the value of 2 chips when 12 chips is the unit.]

   1/4 of 2/3 is 2/12

4. Organizing chips into arrays facilitates fraction multiplication.

5. Delay the use of the multiplication symbol, "x", for a while; use the word "of".

Teaching Actions

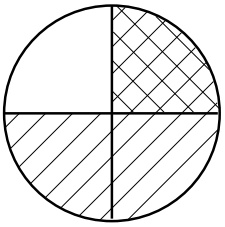
7. Repeat for this problem. Ask students to draw a picture.

   John has a rectangular garden. He is going to plant flowers on half of his garden. He wants 1/3 of the flowers to be roses. What fraction of the garden will be roses?

8. Estimate; solve in groups; share responses; evaluate estimate; record as: "1-third of 1-half is 1-sixth" or 1/3 of 1/2 = 1/6

9. Try another problem:

   Mark, Allie and I ate 3/4 of a pizza. My share was 1/3 of that amount. What fraction of a whole pizza did I eat?



1/3 of 3/4 is 1/4

10. Conclude by asking each group of students to come up with a story problem for 1/2 of 2/3.

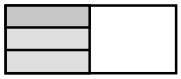
11. Student Page A provides four more story problems for students to work on.

Comments

6. The solution via a picture corresponds to the actions on the chips.

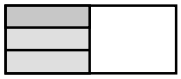
1/2

[Show 1/2 using a continuous model.]



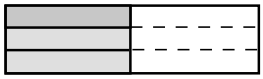
1/3 of 1/2

[Redefine the unit and show 1/3.]



1/3 of 1/2 is 1/6

[Re-evaluate the value of the double-slashed area when the whole rectangle is the unit. Students may need to draw in missing lines.]



Solve problems with chips or pictures. Draw a picture of your solutions. Record a number sentence for each problem.

1. The Martinez family had 2/3 of a sheet cake left over from a birthday party. Today Manuel ate 1/4 of the cake that was left over. How much of the whole cake did he eat?

2. Tom earned $12 babysitting for the Bina family. He put 2/3 in the bank. He spent 1/2 of what was left on admission to the museum. What fraction of the babysitting money did he spend at the museum?

3. The length of a bookshelf is 3/4 of a yard. Jan put her tape deck on 1/3 of the book shelf. What fraction of a yard did her tape deck cover?

4. Study hall lasts 5/6 of an hour. Karen spent 1/2 of her time in study hall on her math lesson. What fraction of an hour did she spend on math?